

Decentralized Finance: Constant Product Automated Market Makers and Liquidity Pools

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ABSTRACT

Decentralized finance (DeFi) platforms rely heavily on automated market makers (AMMs) governed by deterministic mathematical invariants such as $x * y = k$. We model liquidity provider returns, impermanent loss, and arbitrage dynamics across Ethereum decentralized exchanges. Our empirical findings reveal structural trade-offs between capital efficiency and price slippage in volatile cryptocurrency asset markets.

Keywords: Blockchain, Decentralized Finance, Automated Market Maker, Liquidity Pools, Smart Contracts

1. Introduction

Traditional financial exchanges utilize centralized limit order books. In contrast, decentralized exchanges rely on smart contracts holding token reserves in automated liquidity pools, enabling peer-to-pool asset swaps without trusted intermediaries.

2. Mathematical Formulation

The constant product formula maintains the invariant $x * y = k$ where x and y represent asset balances. When liquidity providers deposit paired assets, price changes induce impermanent divergence loss relative to holding the assets outside the pool.

3. Market Efficiency Analysis

Arbitrageurs rapidly synchronize on-chain AMM prices with centralized exchange spot prices. However, high gas fees during network congestion create arbitrage latency and execution slippage.

4. Conclusion

Automated market makers democratize liquidity provision but require novel concentrated liquidity mechanisms to compete with centralized financial market depth.